Mohammad Hasan

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Professional Skills

- **Collaboration & Teamwork**
- **Leadership**
- **Problem Solving**
- **Communication**
- **Data Analytics**
- **Software Testing**

Education

Honours Bachelors of Science | University of Toronto | September 2018 – December 2021

Computer Science Specialist

Program GPA: 3.40/4.00

Relevant Courses: Data Structures & Algorithms, Software Design, Neural Networks & Deep Learning, Artificial Intelligence, Machine Learning, Full-Stack Web Programming, Database Design, Systems Programming, Operating Systems.

Experience

HiGarden – The Guardian App | September – December 2020

- **Collaboration & Teamwork:** Worked in an Agile environment with a team of seven to expand and deploy a 2D planting simulator game for a Markham-based startup whose aim is to integrate the game into Ontario's middle-school syllabi, as well as publish it on several app stores.
- Added several new game levels and using design techniques such as object pooling in **C#** and **Unity**, we managed to decrease the size of the game significantly (by almost 73%).

Rotman Commerce Fintech Association | January 2019

- **Leadership:** Led a team of different backgrounds and majors in a team-based case competition for trading stocks using coding and data analytics.
- Ended up gaining a 87% profit of what we started with, as well as coming top 3 in the competition.

Projects

(more)

Pizza Parlour REST API | Python, Flask

- Created a REST API for a virtual pizza parlour restaurant that completely integrates a server-client implementation while using the Factory Design Pattern and adhering to the SOLID principles.
- **Software Testing:** Involves a set of comprehensive tests that results in 92% coverage, using Python unit-tests and Mocha testing.

Sokoban Solver | Python

- Developed an Artificial Intelligence model to navigate and win the classic game of Sokoban.
- **Problem Solving:** The model is capable of solving several different Sokoban puzzles in less than 5 seconds, with 90% efficiency.

Android Game | Java, UML, Android Studio, SQLite

- **Communication:** Worked with a team of four to create an Android app that has 3 mini-games in it.
- Used various design patterns in the implementation of the app, and used the SQLite database to hold the users' data and assist in authenticating users.

Answer Correctness Predictor | Python, PyTorch, TensorFlow

- **Data Analytics:** Used an ensemble of several algorithms such as K-Nearest Neighbours, Neural Networks, and Item Response Theory, to solve classification problems such as predicting students' answers to Khan Academy's quizzes.
- Our Machine Learning model achieved high accuracy, placing my team and I in the top 10% for a competition.

Hard Skills & Certifications

- Certified **Associate** in **Python** (Python Institute)
- Certified **Data Scientist** (DataCamp)
- Programming in **Python, Java, C, SQL, JavaScript, HTML, CSS, Node.js, React**
- Game Development with **Unity** and **C#**
- Data Science with **Python, Pandas, Seaborn,** and **NumPy**
- Backend Development with **Django, Flask, SQLite**
- Machine Learning with **Python, TensorFlow** and **PyTorch**